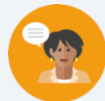


6.5 Drawing Transformations

Lesson Introduction

 Benchmark	MA.912.GR.2.5 Given a geometric figure and a sequence of transformations, draw the transformation on a coordinate plane.		 Learning Target	I can draw the transformed figure when given a geometric figure and a sequence of transformations as a written description or an algebraic description.
 Benchmark Coherence	Building On MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.		Working Toward N/A	
 Essential Questions	How is applying transformations using a written description similar or different from using an algebraic description?			
 Lesson Narrative	This lesson builds on the idea that we can follow and perform a variety of transformations. Students will be asked to apply sequence of transformations to a given preimage by graphing the images using the written and/or algebraic descriptions given.			
 Component Summary	6.5.1 Warm-Up (~5 min.)	Error analysis of a description of a sequence of transformations (<i>SE p. 307</i>)	6.5.3 Guided Practice (5-10 min.)	Applying a sequence of transformations given as a written description (<i>SE p. 309</i>)
	6.5.2 Guided Instruction (25-30 min.)	Applying transformations given as written and algebraic descriptions (<i>SE p. 308</i>)	6.5.4 Your Turn! (5-10 min.)	Applying a sequence of transformations given as an algebraic description (<i>SE p. 310</i>)
	6.5.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 310</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Transformation - Dilation - Rotation - Reflection - Dilation		(Optional) Posters (Optional) Markers	
		Additional Preparation		
		If using the Gallery Walk, posters may need to be made of each question ahead of time.		

 Teacher Tips	Common Misconceptions	Things to Consider
	- Student may apply the rotation in 6.5.2 Guided Instruction around the origin instead of Point F. - Students may apply only two of the three transformations in 6.5.3 Guided Practice.	For all transformations described using algebraic descriptions, consider having students name the transformation that will occur prior to applying the transformation.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Take Turns 6.5.2 Guided Instruction questions 1-3 can be presented as a Gallery Walk. During the work at the posters, students take turns in the activity by choosing a different group member to do all of the writing based on the group's discussion. This move requires all members to participate and listen to the discussion and prevents one student from taking over.	MA.K12.MTR 2.1: Multiple Representations The transformations in 6.5.2 Guided Instruction are stated in word description and in algebraic description. Including both of these allows students to make connections between the concepts and representations in both forms and the coordinate plane.
	In 6.5.3 Guided Practice, students are applying a sequence of three transformations. Consider differentiating the product by eliminating one of the transformations or by breaking the question into two questions, with the second question beginning with the image produced by the first or second transformation.	
Lesson Notes:		

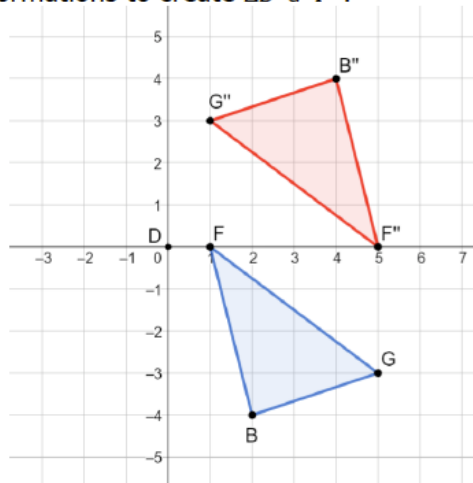
6.5.1 Warm-Up: Error Analysis

Component Overview: Two sample student responses to determine a sequence of transformations applied to a figure are given. Students will determine and justify their reasoning for which response accurately describes the sequence.



6.5.1 Warm-Up: Error Analysis

- $\triangle BGF$ was transformed by a sequence of transformations to create $\triangle B''G''F''$.



This component can provide valuable data for which students still need remediation on describing transformations. Monitor students' responses as they work to determine who may need extra support during the day's lesson.

Brian and Emily were determining the sequence of transformations and wrote their answers in the table.

	$\triangle BGF$ Transformed to $\triangle B'G'F'$	$\triangle B'G'F'$ Transformed to $\triangle B''G''F''$
Brian's Answer	Rotated 180° about the origin	Translated right 6 units
Emily's Answer	Reflected over the x -axis	Translated right 4 units

Determine whose sequence of transformations is correct. Explain your reasoning.

Brian is correct because his sequence of transformations maps $\triangle BGF$ to $\triangle B''G''F''$.

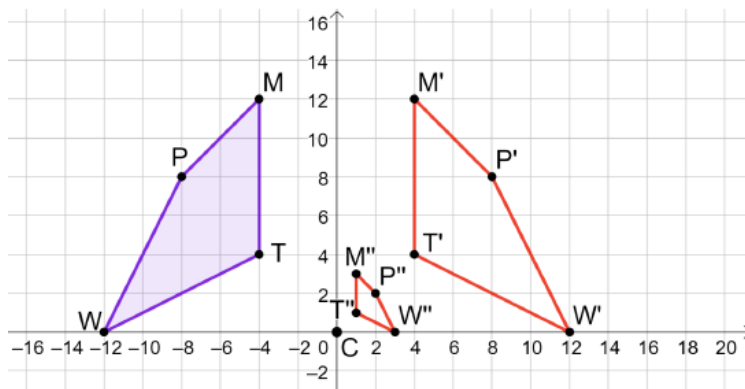
6.5.2 Guided Instruction: Drawing Transformations

Component Overview: Students apply a sequence of transformations to a given figure by graphing the image.



6.5.2 Guided Instruction: Drawing Transformations

1. $MTWP$ and point C are shown on the coordinate plane.



- Draw the quadrilateral that is a reflection of $MTWP$ over the y -axis on the coordinate plane above. Label the quadrilateral $M'T'W'P'$.
Image is drawn on the graph.
- Draw the quadrilateral that is a dilation of $M'T'W'P'$ centered at point C using a scale factor of $\frac{1}{4}$ on the coordinate plane above. Label the quadrilateral $M''T''W''P''$.
The image is drawn on the graph.
- Compare your drawings with your partner. Make any corrections if needed.



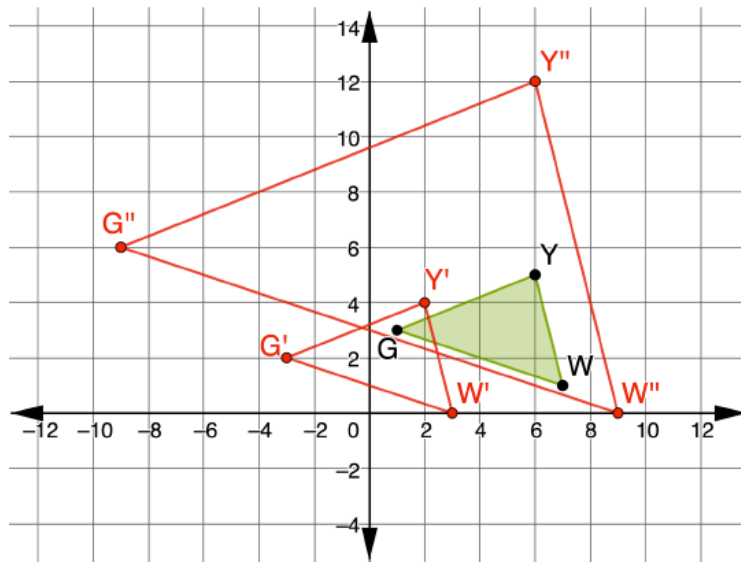
Gallery Walk

Post each question around the room on a separate poster. For large classes, post three different sets of the questions. Place students in groups of two or three. At each poster rotation, utilize the Take Turns routine by having a different group member take the marker and be responsible for any writing or drawing done.

- Each group begins at a poster, discussing and applying the first transformation.
 - If so, apply the second transformation.
 - If not, provide written feedback for why the first transformation was not correct and then draw the correct transformation.
- Groups rotate to the next poster and check that the first transformation was completed correctly.
 - If both transformations were done correctly, write a positive statement about the work or a suggestion of another strategy that could have been used.
 - If any of the work is incorrect, provide written feedback and then draw the correct transformations.
- Groups rotate back to their original poster and review all comments and/or edits.

6.5.2 Guided Instruction: Drawing Transformations (continued)

2. $\triangle GYW$ is shown on the coordinate plane.



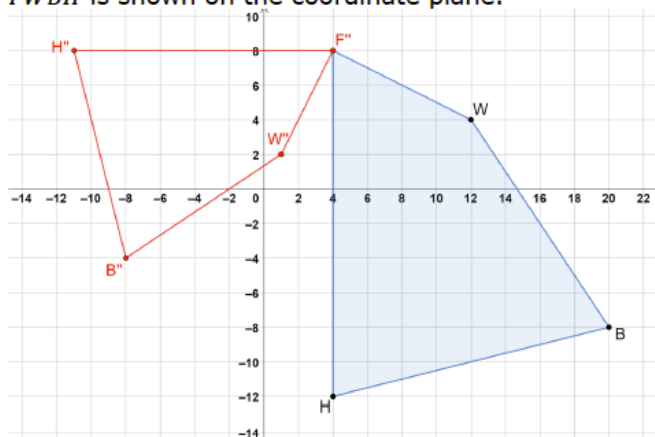
- Draw $\triangle G'Y'W'$ as a result of $(x, y) \rightarrow (x - 4, y - 1)$ on the coordinate plane.
The image is drawn on the graph.
- Draw $\triangle G''Y''W''$ as a result of $(x, y) \rightarrow (3x, 3y)$ on the coordinate plane.
The image is drawn on the graph.



Students may incorrectly apply the dilation as a reduction by dividing by 3 instead of an enlargement by multiplying by 3.

6.5.2 Guided Instruction: Drawing Transformations (continued)

3. $FWBH$ is shown on the coordinate plane.



On the same coordinate plane, draw $F''W''B''H''$, which is a result of dilating $FWBH$ using a scale factor of $\frac{3}{4}$, with the center at point F , then rotating the quadrilateral 90° clockwise about point F .

The image is drawn on the graph.



Students may incorrectly rotate around the origin instead of around point F .

If students use algebraic descriptions for determining the placement of rotated points, ask them why this cannot be used on question 3. Guide students for choosing another method by asking questions like:

- What kind of lines intersect in a 90° angle?
- What do we know about the slope of perpendicular lines?
- How can I use slope and distance to determine where to draw a line segment rotated 90° ?

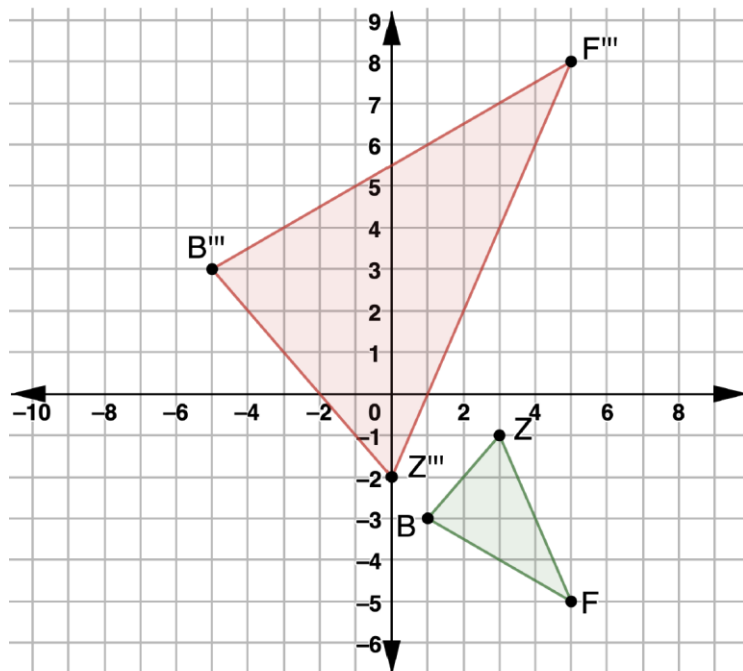
6.5.3 Guided Practice: Sequence of Transformations

Component Overview: Students are challenged to perform a sequence of three transformations.



6.5.3 Guided Practice: Sequence of Transformations

1. $\triangle FZB$ is shown on the coordinate plane.



On the same coordinate plane draw, $\triangle F'''Z'''B'''$, which is the result of a reflection over the x -axis, then translated left 3 units and down 3 units, then dilated with a scale factor of $\frac{5}{2}$ using point Z'' as the center of dilation.

Image is drawn on the graph.



Question 1 is a sequence of three transformations. Students may benefit from working with a partner. Consider breaking students into pairs by using musical pairings. Have students all raise a hand when the music begins and walk randomly around the room. Once the music stops, they high-five the person nearest them who then becomes their partner.



Consider differentiating the question by eliminating one of the transformations or by breaking the question into two questions, with the second question beginning with the image produced by the first or second transformation.

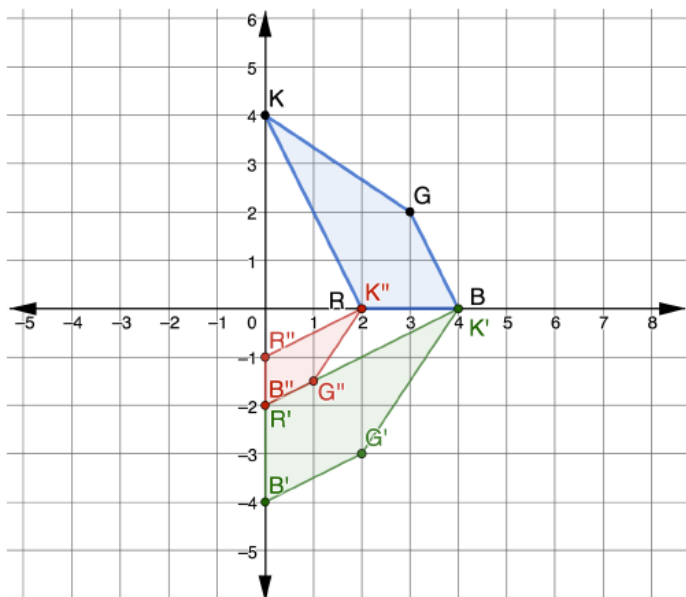
6.5.4 Your Turn!

Component Overview: Students apply a sequence of transformations presented as an algebraic description.



6.5.4 Your Turn!

1. $KGBR$ is shown on the coordinate plane.



On the same coordinate plane, draw $K''G''B''R''$, which is the result of $(x, y) \rightarrow (y, -x)$, then $(x, y) \rightarrow (\frac{1}{2}x, \frac{1}{2}y)$.

Image is drawn on the graph.



Before students begin applying the transformations, encourage them to anticipate what transformations will occur based on the algebraic descriptions.

6.5.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



6.5.5 Wrap-Up: Cool Down

1. Why do you think it is important to use the prime notation when working with transformations?

Answers will vary.